

Deep Watch

Reasoning Under Pressure Beneath the Surface

10 WATCHES · 41 STOPS · GRADE 6 · DEEPWATCH_MS

THE OPENING CARD

You have the watch on a submarine, ninety metres down. Nobody aboard can see out, and neither can you. What you know about the water comes from listening. What you know about the boat comes from people reading gauges to you over a telephone. Tonight something is out there that nobody has named, water is rising inside the hull, and the pencil line on the chart drifts further from the truth every minute. Commander Iris Vance can order the boat up, where all of it stops being deadly. Somebody has to be willing to say so.

Boat Walkdown

BEFORE THE WATCH — GREET

Twelve people in a steel tube, and you know none of them

There is nowhere on this boat you can be alone. The people who fix the leaks and the people who read the gauges work an arm's length apart. Put a name to as many of them as you can now, before something goes wrong and you have to shout one.

PLAN CARD 73W

Two hours to sailing. The fire main valve in the forward compartment is chained shut, and nobody will say who chained it. Chief Ferro wants to walk you through the boat before your first watch tonight. This watch you learn the forward spaces, and how to get from one end of the boat to the other without looking. Tonight you may have to find that valve in the dark, in smoke, by feel.

BRIEFING 42W

You have the watch for the first time. Before anything else happens, you have to be able to find your way around this hull in the dark, and know what is in each space and how you would get out of it.

WHAT YOU DECIDE 13W

Learn the boat as a system of spaces, not a list of names.

1.1 What a walkdown is for

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A ROOM · SEQUENCE

SCENE 30W

Chief Petty Officer Dario Ferro walks you into the forward space and tells you to learn it. Kit down both sides. A hose reel on the wall. Two ways out.

GUIDE 58W

All four of these can be done in the same minute. So this is not a timetable. Ask instead what each fact lets you understand. A locker of props means nothing until you know what this space can do to you. A hatch you have not tried is a guess. One of these decides whose casualty it is.

ASK 12W

Learn a compartment in the order the facts depend on each other.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a submarine is split into compartments that can be sealed off from each other

VERDICT 77W

A submarine survives by being divided. The sealed boundary matters first. Ninety metres down it is what keeps a flooded compartment from becoming a flooded boat. The ways out come next, and they have to be tried rather than assumed. Then look for what could start a casualty in here: sea pipes, cables, stores. Only then does the kit mean anything. A locker is the right locker once you know what this space can do to you.

TAKEAWAY 17W

A compartment is a sealed box first, and everything else you learn about it hangs off that.

1.2 Every fitting is an argument

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A PERSON · PROTOCOL

SCENE 33W

Chief Machinist's Mate Jun Haruki stops at four fittings on one wall and asks what each is for. None has a label, only a number. He steps back from the valves and waits.

GUIDE 52W

None of these four has a label, only a number. So work from the shape, the height, and what is on the far side. Ask of each one what opening it would let in, or let out. The result is the clue. One of them has the sea behind it, pushing hard.

ASK 10W

Match each fitting to what it is there to do.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

water pressure rises by about one atmosphere for every ten metres of depth

VERDICT 75W

A fitting makes sense once you join its shape to what lies behind it. A wheel valve through the hull may have the sea behind it, at pressure. A manifold at head height can feed breathing air. A strapped locker can hold props and plugs. A hose reel can put water on a fire. The point is not to learn four shapes. It is to work out what a thing does from where it sits.

TAKEAWAY 16W

What a fitting is for is decided by what is on the other side of it.

1.3 The trip back without a waypoint

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · ROUTE

SCENE 33W

Lieutenant Imani Sowande, the navigator, sends you aft with a sealed note. On the way back one door will be shut and the marker lights off. You have walked it once, with labels.

GUIDE 72W

First you walk the route with the labels on. Then the labels go away and you walk it back from memory. One door will be shut and the marker lights will be off. When that happens, look for the landmark in the compartment you are actually standing in, and use it to work out where the detour has put you. You are graded on the compartments you reach, not on how fast.

ASK 21W

Run the route from memory. After the shut door, use the landmark to work out where the detour has put you.

BEHIND THE BUTTON 149W

Why places beat counting. A remembered count of turns only works while nothing interrupts it. Miss one turn and every number after it is wrong, and you have no way of noticing. A remembered place is different: two helm wheels means Control whether you arrived by the route you planned or by a detour.

What a landmark is doing on the list. Each compartment on a submarine has something in it that is nowhere else — the wheels, the radio racks, the long table, the pump. Those are how the crew knows where they are in the dark, and they are the same thing you use when the door is shut.

Why the lights go out. Marker lights fail, smoke fills a passage, and power is lost. The route has to survive that, so the second run is the real test and the first one is only the lesson.

TAKES AS READ

a submarine is divided into named compartments · a route can be remembered as a list of places

VERDICT 73W

A count only works while nothing interrupts it. A turn list fails after one extra turn. A route made of places is different because each compartment becomes a new reference point. If you are diverted, identify the space you enter. Then rejoin the sequence from there. Submarines use the same logic when dead reckoning. The useful question is not how many steps have passed. It is what trusted reference remains after the interruption.

TAKEAWAY 14W

A route remembered as places survives an interruption. A count of turns does not.

A compartment is defined by its boundaries, what it holds, and how you leave it.

Contact in the Noise

BEFORE THE WATCH — TRIAL

Walk the boat from bow to stern once

The boat is one line of compartments with a hatch between each pair. Walk the whole line once. A hatch is a small hole, and the drill you do at four in the morning is done in the dark by feel.

PLAN CARD 82W

Second day out. Sonar is listening for sound in the water, and there are four traces on the display. Nils Rask has been arguing with his own operator about the fourth. It holds almost the same bearing, and it is getting louder. Commander Iris Vance wants it classified before the boat commits. This watch you sort out what is out there, and say what a steady bearing means. A bearing that does not change is two ships heading for the same water.

BRIEFING 43W

Sonar has four sources on the display at once. Two of them have been there since before the boat slowed, one came with the turn, and one comes and goes. The captain wants to know which of them she has to care about.

WHAT YOU DECIDE 12W

Work out what is out there before the boat commits to anything.

2.1 Four traces, one of them ours

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · PROTOCOL

SCENE 50W

Four traces sit on the display while the watch calls them out. The boat slowed four minutes ago and one trace changed with it. Sonar is how a submarine sees: it listens, because light does not travel down here. Petty Officer Nils Rask, who runs it, wants each trace named.

GUIDE 57W

Each trace gives you more than one clue. Does its direction hold? Is it one pitch or many? How long does it last? Did it change when the boat changed speed? Pair on all of that, not on which trace is loudest. The boat slowed four minutes ago. That is a test one of these has taken.

ASK 8W

Match each trace to what is making it.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

passive sonar only listens; it sends nothing out

VERDICT 79W

Every trace here has more than one clue on it. A sound that follows our own speed is being made aboard. A narrow tone that drifts slowly in direction fits machinery on another ship holding a course. Short chirps that vanish fit something alive better than a hull. And a direction that barely moves while the sound grows louder is the one to say out loud. You name a trace by combining its clues, not by picking the loudest.

TAKEAWAY 12W

A sound that changes when the boat changes belongs to the boat.

2.2 The trace that will not resolve

SONAR & ACOUSTICS · SONAR ROOM · A PERSON · CHOICE

SCENE 37W

The fourth trace is faint. It has held nearly the same direction for eleven minutes, and it shows no tone at all on the narrow display. The boat has not changed speed or depth since it appeared.

GUIDE 57W

Two facts have to be explained together. The trace is faint, and it has no tone. Ask of each option whether it covers both. A layer can bend sound and make something faint. It does not pick out tones and remove them. Eleven minutes is a long life for a flicker in software. And nothing aboard changed.

ASK 15W

Machinery makes a steady tone. This trace has none. What does that say about it?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

sound bends when it crosses water at a different temperature · a narrow display looks for steady tones, and a broad one measures sound of every pitch at once

VERDICT 75W

The broad display measures sound of every pitch at once. The narrow one hunts for steady tones. A layer of warmer water can bend a path and make a far-off source arrive faint. It cannot reach in and strip that source's tones away. So faintness and missing tones are two separate clues. The layer explains the first. The second says no tone is loud enough to show. Eleven minutes says it is not a flicker.

TAKEAWAY 15W

Bending can make a sound faint. It cannot pick out and remove that sound's tones.

2.3 The bearing that does not move

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · CHOICE

SCENE 33W

The contact has stayed within half a degree of the same bearing for six minutes. It gets louder on every sweep. The boat is still making eight knots, and the channel narrows ahead.

GUIDE 53W

A bearing tells you a direction, not a distance. So nothing here is settled by bearing alone. Ask what each option needs you to already know. A steady bearing does not prove a collision. What it does is remove the comfortable answer, that the contact is sliding safely past. The channel narrows ahead.

ASK 12W

What do a steady bearing and a rising level tell you together?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

passive sonar gives a direction and no distance at all

VERDICT 70W

A bearing is a direction, not a distance. If the direction holds while the sound grows louder, the two tracks are not sliding safely past each other. That is what a closing course looks like from a boat with no range. The steady bearing alone would not prove it. The rising level alone would not either. Together they rule out the comfortable answer, and that is enough to act on.

TAKEAWAY 16W

A bearing that holds while the sound grows is a closing course, not a quiet one.

END OF THE WATCH 15W

A bearing is a measurement; a track is an argument built from several of them.

Position Without a Trusted Fix

BEFORE THE WATCH — FOLLOW

Okonjo walks the sea water line

Sea water comes in at one end and goes out at the other, and it passes through six valves on the way. Okonjo walks the line once. Stay with her, because she stops at every valve you would have to shut.

PLAN CARD 81W

Third day, and the last trusted fix was three and a half hours ago. Imani Sowande has a plot built from course, speed and time. Anya Petrov has been reading the depth sounder, and the two do not agree. There is a half-knot current nobody has measured today. This watch you work out how far the water has carried the boat, and get a position that does not depend on the plot. A bank sits half a mile off the track.

BRIEFING 42W

The boat has been submerged for hours on dead reckoning. The plot is a confident-looking line drawn from course and speed, and it knows nothing about the water that has been setting the boat sideways the whole time. Ahead is a bank.

WHAT YOU DECIDE 13W

Work out not where the boat is, but how well that is known.

3.1 How far the water has carried us

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · BALLPARK

SCENE 31W

The last trusted fix was three and a half hours ago. The current here runs about half a knot. Nobody has put it on the plot, because nobody has measured it.

GUIDE 59W

Five numbers, and three of them belong to the boat: our speed, our depth, our course. Ask of each whether it moves the boat through the water, or moves the water itself. Our own speed cannot see a current. Half a knot for three and a half hours is a long way. This is the plot for the narrows.

ASK 13W

How far could the boat be from where the plot says it is?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the log measures speed through the water, not speed over the ground

VERDICT 78W

Between fixes the boat carries its position forward from the last known point, using course, speed and time. That is dead reckoning. The water moves as well. A current can carry the whole boat sideways, and it never shows up in the speed through the water. Half a knot for three and a half hours is nearly two sea miles. The error grows every hour nobody measures it. This is the plot the boat enters the narrows on.

TAKEAWAY 13W

Dead reckoning tells you where you would be if nothing had pushed you.

3.2 Two displays, one source

NAVIGATION & POSITION • CONTROL ROOM • A PERSON • BALLPARK

SCENE 38W

The two navigation displays agree to within a hundred yards. The sounder's ping comes back in 0.120 seconds. The chart says one hundred and two metres here. The last trusted fix was three and a half hours ago.

GUIDE 62W

Work the sounder out yourself. The ping goes down and comes back, so the time covers the trip twice. Sound travels about fifteen hundred metres a second in water. Pick the speed, the time and the number you divide by, and see whether the depth matches the chart. One tile is the chart's own figure, which is not part of the sum.

ASK 13W

How deep is the water under the boat, from the sounder's own ping?

BEHIND THE BUTTON 137W

Why the two displays are one display. Both take their position from the same source. When the source is out, both are out by the same amount, and they still agree. Agreeing is only worth something when the second reading could have gone wrong in a different way.

Why the echo time is halved. The sound has to get down to the bottom and back up again before the sounder hears it. The time on the screen covers both trips, so the depth is half of what the sound travelled.

What a disagreement is for. Ninety metres against a charted one hundred and two does not tell you which one is wrong. It tells you that something needs checking before the boat runs into banked water, and that the check has to come from somewhere else again.

TAKES AS READ

an echo time covers the trip down and the trip back • two displays fed by one source can be wrong in the same way

VERDICT 78W

The sounder measures a time. Sound goes down and comes back at about fifteen hundred metres a second, so 0.120 seconds is one hundred and eighty metres of travelling. The bottom is half of that: ninety metres. The chart says one hundred and two, so something is wrong by twelve. The two displays cannot settle it, because both repeat one source. That is one measurement shown twice. The sounder is the only reading here that could have disagreed.

TAKEAWAY 13W

Agreement is only a check when the second measurement could have failed differently.

3.3 Where the sound will not go

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · CHOICE

SCENE 34W

The sound trace reads faster above sixty metres and slower below it. A contact held for forty minutes while the boat stayed at twenty. Then it vanished in one sweep, and nothing else changed.

GUIDE 59W

The contact went in one sweep, and nothing else changed. Ask of each option whether it explains that suddenness. Distance and shutting down both work slowly. The blind sector is a direction, not an event. The sound speeds are the numbers doing the work. Sound bends toward slower water, and that can leave a gap our sound never enters.

ASK 14W

The contact was held near the surface and is now gone. Where is it?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

sound travels at different speeds through warmer and colder water · sound bends toward the slower water

VERDICT 79W

Sound does not travel in straight lines through the sea. It bends toward the slower water, and the water below sixty metres here is slower. Bend a path far enough and there are places the sound from our own depth never reaches. A contact that crosses into one of those does not fade away. It goes in a single sweep, while every other trace stays where it was. That suddenness is the shape of a shadow, not of distance.

TAKEAWAY 16W

Sound bends as it travels, which leaves places a listening boat cannot hear into at all.

3.4 What a walkdown is for

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A ROOM · CALLBACK · SEQUENCE

SCENE 30W

Chief Petty Officer Dario Ferro walks you into the forward space and tells you to learn it. Kit down both sides. A hose reel on the wall. Two ways out.

GUIDE 58W

All four of these can be done in the same minute. So this is not a timetable. Ask instead what each fact lets you understand. A locker of props means nothing until you know what this space can do to you. A hatch you have not tried is a guess. One of these decides whose casualty it is.

ASK 12W

Learn a compartment in the order the facts depend on each other.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a submarine is split into compartments that can be sealed off from each other

VERDICT 77W

A submarine survives by being divided. The sealed boundary matters first. Ninety metres down it is what keeps a flooded compartment from becoming a flooded boat. The ways out come next, and they have to be tried rather than assumed. Then look for what could start a casualty in here: sea pipes, cables, stores. Only then does the kit mean anything. A locker is the right locker once you know what this space can do to you.

TAKEAWAY 17W

A compartment is a sealed box first, and everything else you learn about it hangs off that.

END OF THE WATCH 13W

Precision is not accuracy, and two displays sharing a source are one measurement.

Silent Passage

PLAN CARD 73W

Fifth day. The narrows are four miles across, the passage takes eleven hours, and the plot's doubt is a circle two miles wide. Lieutenant Commander Theo Whitfield, second in command, wants to wait for a better fix. Vance points out that waiting puts the boat mid-strait at first light, which is worse. This watch you decide what has to be true before the boat commits, and how much margin the passage really has.

BRIEFING 30W

The route ahead narrows between the bank and a shipping lane. Command wants the boat through it with contact held throughout and as little acoustic exposure as the plant allows.

WHAT YOU DECIDE 12W

Cross a constrained area without losing contact awareness or making unnecessary noise.

4.1 What has to be true before the boat commits

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · SEQUENCE

SCENE 33W

Rask shuts the sonar-room door and lays out the checks for the passage. The narrows are on the next chart panel. Once the boat is committed there is no room to turn round.

GUIDE 61W

Each of these four makes the next one readable. So ask what would make no sense without it. A new sound is only new against a record of the old ones. A missing tone means nothing until you know where the layer is. Watching a contact move takes minutes. And one of these is agreed early, because afterwards everybody is busy.

ASK 12W

Put the passage checks in the order they have to be done.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a measurement can mean nothing without an earlier one to read it against

VERDICT 79W

A plan is a list of things that have to be true when the boat commits. Each one is what the next gets read against. Without the boat's own noise, a new sound is only a sound. Without the layer, a missing tone looks like an empty sea instead of a bent path. Bearings take time, so they cannot be started at the entrance. And the lost-contact rule is agreed early for a different reason. Afterwards, everybody is busy.

TAKEAWAY 14W

A plan is ordered by what each measurement needs before it can mean anything.

4.2 The margin that survives being wrong

NAVIGATION & POSITION · CONTROL ROOM · A PERSON · BALLPARK

SCENE 30W

The narrows fill the next chart panel. They are four sea miles across, the bank is on the left, and the circle round our own position is two miles wide.

GUIDE 62W

Four numbers, and one is our own speed, which does not belong in a clearance. Another is the number of halves, which is a step and not a measurement. The circle round our position is how wide the doubt really is. That whole circle has to fit in safe water. Four miles sounds like plenty until the doubt is drawn on it.

ASK 15W

How much clear water is left if the boat runs the middle of the channel?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a plotted position is the middle of a circle of doubt, not a fact

VERDICT 74W

The circle round a plotted position is the honest width of where the boat might be. So a route is planned against the circle, not against the line at its middle. The whole circle has to stay in safe water. Half the channel is two miles. Half the circle is one mile. What is left over is one mile. Four miles of channel sounds generous until the doubt is drawn at the same scale.

TAKEAWAY 12W

A route is chosen against the doubt, not against the pencil line.

4.3 The one shaft speed that rings

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · CHOICE

SCENE 29W

At ninety-two turns the after compartment rings loudly enough to hear through a bulkhead. At eighty-five and at a hundred it does not. Nothing else in the plant changes.

GUIDE 65W

All four options are real faults, and three of them get worse as speed goes up. Ask of each whether it could be loud at one speed and quiet either side. Balance problems grow. Bearings get worse. Bubbles start and stay. Only something that has to match a rhythm behaves like this. That decides whether the fix is a repair or a speed to avoid.

ASK 13W

The hull rings at one shaft speed and nowhere else. Why that one?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

everything has speeds at which it prefers to shake, the way a swing does

VERDICT 72W

Everything has a rhythm it prefers to shake at, the way a swing does. Push a swing at any old moment and little happens. Push it in time and it goes higher on every push. The propeller blades hit the water a set number of times each turn. At one shaft speed that beat lands in time with the compartment's own rhythm. Either side of it, the pushes fall out of step.

TAKEAWAY 14W

A small push repeated in time with something's own rhythm builds a large movement.

4.4 What a walkdown is for

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A ROOM · CALLBACK · SEQUENCE

SCENE 30W

Chief Petty Officer Dario Ferro walks you into the forward space and tells you to learn it. Kit down both sides. A hose reel on the wall. Two ways out.

GUIDE 58W

All four of these can be done in the same minute. So this is not a timetable. Ask instead what each fact lets you understand. A locker of props means nothing until you know what this space can do to you. A hatch you have not tried is a guess. One of these decides whose casualty it is.

ASK 12W

Learn a compartment in the order the facts depend on each other.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a submarine is split into compartments that can be sealed off from each other

VERDICT 77W

A submarine survives by being divided. The sealed boundary matters first. Ninety metres down it is what keeps a flooded compartment from becoming a flooded boat. The ways out come next, and they have to be tried rather than assumed. Then look for what could start a casualty in here: sea pipes, cables, stores. Only then does the kit mean anything. A locker is the right locker once you know what this space can do to you.

TAKEAWAY 17W

A compartment is a sealed box first, and everything else you learn about it hangs off that.

END OF THE WATCH 15W

Quiet costs capability, and the trade has to be made deliberately rather than by default.

Forward Flooding

BEFORE THE WATCH — HUNT

Six breathing sets, and the board finds four

Every compartment is meant to hold a breathing set, and two are not where the board says. A set nobody can find is a set nobody breathes from. Find them before the drill, not during it.

PLAN CARD 70W

Sixth day. Water is over the bilge in the forward space, with the drain pump running. Ruth Hallam has the level and reports it steady. Ferro does not believe steady means anything, and is looking for the hole. This watch you find where the water is coming in, and work out whether pumping can win. Pumping buys minutes, and they are worth nothing unless somebody spends them finding the source.

BRIEFING 41W

The forward space is taking water. The drain pump is running and the level is still rising. Somewhere under those deck plates is a hole, and the boat is at 90 metres, which is what makes it a hole worth respecting.

WHAT YOU DECIDE 12W

Find the water, decide whether pumping can win, and stop the source.

5.1 Under the deck plates

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A ROOM · BALANCE

SCENE 34W

The forward bilge alarm has gone twice, and water shows through the deck-plate slots. Machinist's Mate Ruth Hallam marks the level a minute apart while the drain pump runs. Command wants the inflow now.

GUIDE 63W

Reading a stream costs nothing, so read them all. Then count only the ones in the same units as the number command wants, which is gallons a minute of water coming in. The bilge level is rising while the drain pump is already taking water out, so the pump has to go back into the sum. Report the total when the ledger closes.

ASK 14W

How fast is water coming in, before the pump takes any of it out?

BEHIND THE BUTTON 170W

Why the rise is not the inflow. What piles up in the bilge is what came in minus what has already been removed. The pump is running the whole time Hallam is marking the level, so the rise you can see is the inflow with the pump's share already subtracted. Add the pump back and you have what is actually coming in.

Why the units matter. A centimetre of rise is not a flow until you know the area of the compartment floor and the time between the two marks. Some of the streams on the board are already in gallons a minute and some are not, and adding a number in the wrong units gives an answer that is wrong by a factor nobody can see.

Why command wants the inflow and not the level. The level tells you how bad it is now. The inflow tells you how long you have and whether the pumps you own can win. That is the number a decision is made on.

TAKES AS READ

a rise in the bilge is what stayed behind, not what came in · a pump that is running takes water out during the same minute

VERDICT 77W

The rise in the bilge is only what stayed behind during that minute. The pump ran the whole time, and what it threw over the side never showed on the deck plates. So the sum is what piled up plus what was pumped. Eighty-eight and fifty-five make about a hundred and forty-three gallons a minute. The fresh-water tank and the cooling flow did not change, so they add nothing. A pump makes a bad leak look small.

TAKEAWAY 23W

What piles up is what came in minus what was taken out, so anything already removing has to go back into the sum.

5.2 What ninety metres puts behind the valve

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A PERSON · BALLPARK

SCENE 29W

The suspect hull fitting is ninety metres down. Ferro wants to know what the sea is pushing with on the far side of it, before anybody touches that valve.

GUIDE 59W

Five numbers, and two of them belong to a different question. One is how fast the bilge level rose. One is what the pump can shift. Ask of each whether it is about the sea outside or the water inside. Pressure comes from depth, and nothing else here. What is behind the valve is what decides who opens it.

ASK 11W

About how much pressure is the sea putting on the fitting?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

water pressure rises by about one atmosphere for every ten metres of depth

VERDICT 80W

Water is heavy, and every metre of it above the fitting presses down. About ten metres of sea adds one atmosphere, so ninety metres adds nine. Add the air we all live under and the outside of that fitting sits near ten atmospheres. That is why a small hole this deep drives so much water. It is also why shutting the hull valve matters more than fitting another pump. A pump fights the flow. The valve takes the push away.

TAKEAWAY 15W

Pressure grows straight in step with depth, so depth decides how hard a leak pushes.

5.3 What else that valve feeds

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · PROTOCOL

SCENE 32W

Two valves will stop the water: the one at the hull and the one inboard. The same header feeds the sonar cooling. Haruki wants each order matched to what it really does.

GUIDE 62W

Four orders, and every one takes something from somebody. The same pipe that feeds the leak cools the sonar. Ask of each order which side of the hull it acts on, and what stops as a result. Only one cut separates the boat from the sea. Two of the others change how fast the water comes in, not whether it comes in.

ASK 8W

Match each order to what it actually does.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

one pipe header can serve more than one system

VERDICT 79W

Shutting a valve is never free, because a pipe rarely serves one thing. Here the same header carries the leak and the cooling for the sonar array. Only the valve at the hull separates the boat from the sea, so only that one ends the flood. The inboard valve stops the flow at the break and leaves the header full. A pump buys time. Telling sonar moves no water, and it is why that watch is not caught out.

TAKEAWAY 14W

Every valve you shut takes something away from somebody, and the order decides what.

5.4 The trip back without a waypoint

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · CALLBACK · ROUTE

SCENE 33W

Lieutenant Imani Sowande, the navigator, sends you aft with a sealed note. On the way back one door will be shut and the marker lights off. You have walked it once, with labels.

GUIDE 72W

First you walk the route with the labels on. Then the labels go away and you walk it back from memory. One door will be shut and the marker lights will be off. When that happens, look for the landmark in the compartment you are actually standing in, and use it to work out where the detour has put you. You are graded on the compartments you reach, not on how fast.

ASK 21W

Run the route from memory. After the shut door, use the landmark to work out where the detour has put you.

BEHIND THE BUTTON 149W

Why places beat counting. A remembered count of turns only works while nothing interrupts it. Miss one turn and every number after it is wrong, and you have no way of noticing. A remembered place is different: two helm wheels means Control whether you arrived by the route you planned or by a detour.

What a landmark is doing on the list. Each compartment on a submarine has something in it that is nowhere else — the wheels, the radio racks, the long table, the pump. Those are how the crew knows where they are in the dark, and they are the same thing you use when the door is shut.

Why the lights go out. Marker lights fail, smoke fills a passage, and power is lost. The route has to survive that, so the second run is the real test and the first one is only the lesson.

TAKES AS READ

a submarine is divided into named compartments · a route can be remembered as a list of places

VERDICT 73W

A count only works while nothing interrupts it. A turn list fails after one extra turn. A route made of places is different because each compartment becomes a new reference point. If you are diverted, identify the space you enter. Then rejoin the sequence from there. Submarines use the same logic when dead reckoning. The useful question is not how many steps have passed. It is what trusted reference remains after the interruption.

TAKEAWAY 14W

A route remembered as places survives an interruption. A count of turns does not.

Flooding is stopped at the hull fitting; everything else is delay.

Electrical Fire

PLAN CARD 71W

Same day, eight hours after the flooding, and nobody has slept. Smoke is coming off a switchboard in the electrical space. The indicator says the bus is open, and a meter on the cable inside reads 440 volts. Bede Okonkwo will not send anybody in on the strength of a lamp. This watch you decide what that meter is telling you, and how to fight a space nobody can see into.

BRIEFING 30W

Smoke in the electrical distribution space. The switchboard indicator says the affected bus is open, and the fire is behind a panel where nobody can see the seat of it.

WHAT YOU DECIDE 17W

Put out a fire whose ignition source is electrical, in the order that makes it stay out.

6.1 Three legs, three different actions

FIRE & ELECTRICAL · ELECTRICAL DISTRIBUTION · A ROOM · PROTOCOL

SCENE 30W

Chief Electrician's Mate Bede Okonkwo puts the three legs of the fire on the board. The fault that keeps lighting it. The insulation that burns. The air in the compartment.

GUIDE 64W

Three of these are legs of the fire. One is a warning about the space next door. Pair each by asking what it gives the fire: heat, fuel, or air. Taking a leg away is not the same as watching a symptom. A dead cable can still be hot enough to light again. So proving it dead and cooling it are two different jobs.

ASK 12W

Match each leg of the fire to the action that removes it.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

a fire needs fuel, air and something to start it

VERDICT 75W

A live fault keeps pouring heat into insulation that is already alight. So the first move is to open the breaker and prove the cable dead. That stops the heating and the shock together. It does not stop the fire. Hot insulation burns on until something cools it. The air is then controlled, to starve the flame and steer the smoke. A hot bulkhead is a warning to watch, not a fourth thing to fight.

TAKEAWAY 18W

Killing the power stops the heating and the shock. It does not put out what is already burning.

6.2 What a lamp is worth

FIRE & ELECTRICAL · ELECTRICAL DISTRIBUTION · A PERSON · SEQUENCE

SCENE 33W

Smoke is pushing through the door. The switchboard lamp says the bus is open, and a meter on the cable reads four hundred and forty volts. The hose team is masked up, waiting.

GUIDE 58W

The lamp and the meter disagree, and the order has to survive that. Ask of each step what it proves, not what it believes. A lamp reports what the switchboard thinks. A meter reports the cable. Water on a live cable carries the current back to the hose team. They are masked up and waiting on this order.

ASK 13W

Put the steps in the order that proves it before anybody fights it.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a bus can be fed from more than one source

VERDICT 73W

The lamp reports what the switchboard believes. A meter on the cable reports what the cable is doing. The two can disagree, because contacts stick, lamps fail, and another source can feed the same cable from behind. Water on a live cable carries current back into the hose team. So the order is fixed. Open the source, test the cable, prove it dead, then fight. The meter turns a belief into a fact.

TAKEAWAY 12W

An indicator reports what somebody believes. A meter reports what is there.

6.3 Smoke is not only a breathing problem

ATMOSPHERE & LIFE SUPPORT · AUXILIARY MACHINERY & BILGE · A ROOM · CHOICE

SCENE 34W

You cannot see half a metre. The far bulkhead is eight metres away. Head height reads sixty degrees against twenty-eight at the deck. The fire is behind a panel where nobody can see it.

GUIDE 59W

All four options are true of a smoky space in some way. They differ in what the smoke took away. Masks fix breathing and nothing else. Thirty degrees between the deck and head height means the heat is in layers. So it can get worse up high while the floor stays cool. Nobody is going to see this fire.

ASK 11W

What does the smoke change about how this casualty is fought?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a fire inside a structure does not show you where it is

VERDICT 78W

Smoke does more than make breathing hard. It takes sight away, so a team can lose the route, the boundary and each other within a few metres. Sixty degrees at head height against twenty-eight at the deck also says the heat is stacked in layers. Things can get much worse above that layer while the floor still feels survivable. Masks fix the breathing and nothing else. From here the space has to be measured rather than looked at.

TAKEAWAY 14W

Smoke takes away the instrument you were using most, which is your own eyes.

6.4 Ours, or out there

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · BELT

SCENE 28W

The sounds held overnight have to be logged before the watch changes. Some of them were made by this boat. The rest came from the water outside it.

GUIDE 37W

Two bins. Turn the boat and watch what happens. A sound made on the boat keeps the same bearing, because it turns with us. A sound from outside keeps its own bearing while we turn under it.

ASK 12W

Send each sound to the bin that says where it was made.

BEHIND THE BUTTON 73W

Why our own noise is hard. It arrives on the same array as everything else, and it can be louder than a real contact. What gives it away is that it does not move when we do.

Why the log matters. Our own noise written down as a contact is a ghost the next watch has to chase. A real contact written down as our own noise is a boat nobody is tracking.

TAKES AS READ

a listening array hears the boat it is fitted to as well as the sea

VERDICT 156W

The array cannot tell you where a sound was made. The geometry can. A sound made on this boat arrives from the same direction whatever we do, so its bearing follows the hull round a turn. A sound made out in the water keeps its own bearing while the boat turns underneath it. That is why a thirty-degree course change is the standard test, and why it is worth the time. There is a second clue. Our own noise starts and stops with something the crew did: a pump cut in, a shaft order, a hatch shut. A whale or another ship has no reason to do anything at the same moment as our engine room. Getting it wrong costs both ways. A ghost in the log wastes the next watch. A real contact filed as our own noise is a boat nobody is watching, on a night when we are trying not to be heard.

TAKEAWAY 12W

A sound that turns with the boat was made by the boat.

6.5 Every fitting is an argument

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · CALLBACK · PROTOCOL

SCENE 33W

Chief Machinist's Mate Jun Haruki stops at four fittings on one wall and asks what each is for. None has a label, only a number. He steps back from the valves and waits.

GUIDE 52W

None of these four has a label, only a number. So work from the shape, the height, and what is on the far side. Ask of each one what opening it would let in, or let out. The result is the clue. One of them has the sea behind it, pushing hard.

ASK 10W

Match each fitting to what it is there to do.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

water pressure rises by about one atmosphere for every ten metres of depth

VERDICT 75W

A fitting makes sense once you join its shape to what lies behind it. A wheel valve through the hull may have the sea behind it, at pressure. A manifold at head height can feed breathing air. A strapped locker can hold props and plugs. A hose reel can put water on a fire. The point is not to learn four shapes. It is to work out what a thing does from where it sits.

TAKEAWAY 16W

What a fitting is for is decided by what is on the other side of it.

END OF THE WATCH 15W

An electrical fire is put out by de-energizing it; the agent only finishes the job.

Atmosphere Degradation

BEFORE THE WATCH — CANVASS

Who heard the noise, and when

Somebody heard a knocking sound on the night watch. Where it came from decides which pump gets taken apart. Ask along the boat until enough of the watch have told you what they heard.

PLAN CARD 69W

Eighth day. The scrubber is running and its outlet reads clean. Elena Navarro has been round berthing with a handheld meter, and it reads far higher than the sensor on that bulkhead. Nine people are asleep in there. This watch you decide whether it is the air or the instrument that is wrong, and how long that space is safe. The crew notice this one before the instruments do.

BRIEFING 34W

Two of the crew in berthing have headaches. The atmosphere plant says everything is normal. Both of those statements can be true at once, and working out how is the whole of this watch.

WHAT YOU DECIDE 14W

Tell a failing sensor from failing air, and find where the air stopped moving.

7.1 The compartment the scrubber cannot reach

ATMOSPHERE & LIFE SUPPORT · AUXILIARY MACHINERY & BILGE · A ROOM · CHOICE

SCENE 29W

The scrubber is running and its outlet reads clean. A handheld meter in berthing reads far higher than the sensor fitted there. Control and machinery read normal on handhelds.

GUIDE 59W

Four readings have to be explained at once. A clean outlet. A high handheld in berthing. A low fitted sensor there. Normal handhelds everywhere else. Ask of each option how many of the four it covers. One fault does not have to explain everything when the evidence points to two. A scrubber can only clean air that reaches it.

ASK 7W

Which explanation accounts for every reading here?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a machine can only clean the air that reaches it

VERDICT 79W

A scrubber can only clean air that reaches it. Shut one damper and a compartment drops out of the loop while the plant's own outlet still reads clean. Here the handheld in berthing agrees with how the people in there feel, and handhelds elsewhere read normal. So the problem is in one space. The sensor fitted in that space disagrees with both, which makes it a second fault. Two faults at once is allowed. The evidence decides, not tidiness.

TAKEAWAY 14W

Which compartments a plant is really treating is decided by where the air goes.

7.2 Nine people and a shut damper

ATMOSPHERE & LIFE SUPPORT · AUXILIARY MACHINERY & BILGE · A PERSON · BALLPARK

SCENE 23W

Nine people sleep in a space with the damper shut. The corpsman, Elena Navarro, wants to know how long that air has left.

GUIDE 57W

Five numbers, and two of them have already been used. The room size and the limit were used to work out how many litres you can allow. Ask which numbers the rate needs. Nine people breathing out is not the same as one. A deadline in hours is something the boat can plan around. Soon is not.

ASK 13W

How long until the carbon dioxide in there rises by one per cent?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a person breathes out carbon dioxide steadily, hour after hour

VERDICT 76W

Anything arriving at a steady rate piles up in step with time. One per cent of that space is three hundred and sixty litres of carbon dioxide. Nine people put out about one hundred and eighty litres between them every hour. Three hundred and sixty divided by one hundred and eighty is two hours. It is a rough number, and it does what the word stuffy cannot. It gives the corpsman a time to work against.

TAKEAWAY 8W

A rate turns a limit into a deadline.

7.3 What one fan takes with it

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · CHOICE

SCENE 31W

The drawing shows one fan feeding the scrubber and the after spaces through the same trunk. The watch bill lists them as two systems, and the air plan counts on both.

GUIDE 57W

All four options treat one fan as a real limit. They differ in what they do about it. Ask of each whether it adds another way to get the air. Running the scrubber harder does not add a path. Sealing the after spaces removes a need instead. Two lines on a watch bill are not two systems.

ASK 16W

One fan feeds the scrubber and the after spaces. What does that mean for the plan?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a system is only as separate as the parts it does not share

VERDICT 73W

One fan can do two jobs and still be one failure. If the scrubber and the after spaces share that fan and that trunk, losing it takes both away at once. Writing them in two boxes on the watch bill does not make them separate. Separate means a different fan, a different supply, or a different route. Until one of those exists, the air plan has a single failure that reaches several spaces.

TAKEAWAY 14W

Two systems sharing one part are one system as soon as that part fails.

7.4 Sixty metres, while the boat works

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A PERSON · HOLD

SCENE 32W

The boat has to hold sixty metres while the galley pumps out, the tubes are flooded for a drill, and the water outside gets colder. The trim pump answers all of it.

GUIDE 44W

Hold the depth inside the band on the gauge. The band gets narrower later, because the boat is quieter and the sonar picture is better when it is not wandering. Each change in weight keeps pushing until you set the pump to match it.

ASK 11W

Hold sixty metres while the boat and the water both change.

BEHIND THE BUTTON 70W

Why the boat drifts. A submarine at depth is balanced. Its weight matches the weight of the water it pushes aside. Pump a tonne over the side and the boat is a tonne light, so it rises.

Why the sea moves you too. Colder water is heavier for the same volume, so the same boat floats up in it. Nothing about the boat changed, and the gauge looks the same.

TAKES AS READ

a submarine is balanced between its weight and the water it pushes aside

VERDICT 137W

A submarine at depth is balanced between what it weighs and the weight of water it pushes aside. Everything in this passage upsets one side of that balance. Pumping the galley over the side makes the boat lighter, so it rises and keeps rising. Flooding the tubes makes it heavier, so it sinks and keeps sinking. Crossing onto the shelf changes neither: the water gets heavier for its size, the boat pushes aside more weight than before, and it floats up exactly as if somebody had pumped. All three keep going until the trim pump is set to match, which is why it is set and left rather than tapped. The band gets narrower under the layer because a boat holding depth is a quiet boat, and a quiet boat is the whole point of being there.

TAKEAWAY 12W

A boat that has changed weight keeps moving until somebody answers it.

7.5 Four traces, one of them ours

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · CALLBACK · PROTOCOL

SCENE 50W

Four traces sit on the display while the watch calls them out. The boat slowed four minutes ago and one trace changed with it. Sonar is how a submarine sees: it listens, because light does not travel down here. Petty Officer Nils Rask, who runs it, wants each trace named.

GUIDE 57W

Each trace gives you more than one clue. Does its direction hold? Is it one pitch or many? How long does it last? Did it change when the boat changed speed? Pair on all of that, not on which trace is loudest. The boat slowed four minutes ago. That is a test one of these has taken.

ASK 8W

Match each trace to what is making it.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

passive sonar only listens; it sends nothing out

VERDICT 79W

Every trace here has more than one clue on it. A sound that follows our own speed is being made aboard. A narrow tone that drifts slowly in direction fits machinery on another ship holding a course. Short chirps that vanish fit something alive better than a hull. And a direction that barely moves while the sound grows louder is the one to say out loud. You name a trace by combining its clues, not by picking the loudest.

TAKEAWAY 12W

A sound that changes when the boat changes belongs to the boat.

A plant reading is a report about the plant, not about the compartment you are standing in.

Compound Casualty

BEFORE THE WATCH — EVADE

Keep clear of the sonar watch

The sonar room is listening on a narrow band, and a hatch banged in the passage outside is a noise in their ears. Get down the boat without going past them, and leave them to listen.

PLAN CARD 79W

Ninth day, and four things at once. Water forward. A bus lost aft. The plot gone soft with the power. And Tunde Adeyemi, who runs the propulsion plant, in the passage with a broken wrist. Vance wants to hold the ordered speed. Theo Whitfield says a boat with four casualties at eight knots, nine miles off a bank, is about to have five. This watch you decide what you keep, what you hand over, and whether the boat slows.

BRIEFING 37W

Flooding forward, a partial power loss, a degraded plot and one injured crewman, all inside four minutes. There are not enough people to do everything, so the first decision is which things do not get done yet.

WHAT YOU DECIDE 12W

Prioritise and delegate when there are more casualties than there are people.

8.1 Four casualties, one crew

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A ROOM · TRIAGE

SCENE 32W

Four reports hit control almost together. Water forward. A bus lost aft. The plot gone soft. A crewman down with a broken wrist. Every team leader is waiting for the first move.

GUIDE 64W

All four are real and all four need somebody. What separates them is how fast each one changes while you deal with the rest. Ask of each whether it grows on its own. Ask whether waiting makes it permanent. A broken wrist hurts and does not spread. The boat is still heading for shallow water, and that clock is not the fastest one here.

ASK 5W

What gets the first team?

BEHIND THE BUTTON 172W

Why first is a different question from best. When there is more than one call on the same hour, the value of doing something now is not how much it matters. It is how much is lost by doing it later. Something important that will be just as fixable in an hour can wait. Something small that will not be fixable can not.

What to look for in the options. Ask which one changes if you wait — a reading that is still moving, a person who is getting worse, a window that closes. Then ask which ones can only be done after something else. Those two questions usually leave one option standing.

Why only one answer is marked. Real shifts do all of these, in some order. The stop asks for the front of that order because that is the only part where the reasoning shows. Read the verdict afterwards even when you are right — it says what the others were waiting on, which is the rest of the answer.

TAKES AS READ

some problems get worse on their own, and some do not

VERDICT 80W

Triage sorts by speed, and by what cannot be undone. It does not sort by what sounds worst. Water coming in takes away the thing holding the boat up, and it spreads into new spaces. A lost bus has already done its damage. A soft plot gets worse with every mile toward shallow water, and slowing down freezes it. A broken wrist hurts and stays the same size for an hour. The first team goes where waiting costs the most.

TAKEAWAY 14W

Order casualties by how fast they get worse, not by how loud they are.

8.2 Who does the things you are not doing

FIRE & ELECTRICAL · ELECTRICAL DISTRIBUTION · A PERSON · DELEGATE

SCENE 33W

The flooding is losing now: the pump moves sixty litres a minute against forty coming in. A fire boundary aft is heating. The air is steady, and a crewman has a broken wrist.

GUIDE 58W

Three problems, and you keep one. For each of the others, pick somebody who can take a real first action, and say what reading or event brings it back to you. An owner with no first action will come back for orders at the worst moment. Then take the watch on the one that needs your own judgement.

ASK 13W

Which problem do you keep, and what does each of the others get?

BEHIND THE BUTTON 164W

Which one to keep. The flooding is already winning: the pump moves sixty litres a minute against forty coming in. The fire boundary aft is getting hotter, so it is the one changing for the worse. The broken wrist hurts and is not getting worse. Keeping the easy job is the mistake, because it is the one that finishes and the other two are the ones that move.

What a first action is. Not "keep an eye on it". Something the owner can do without asking you: log the bilge level every five minutes, take a hand-held reading at the boundary, put a splint on. A handover made of instructions to think is not a handover.

Why every handover needs a way back. You are not giving the problem away — you are lending it. Saying what number brings it back means the owner can hold it without checking in, and you find out at the moment it matters rather than at the end.

TAKES AS READ

how fast a problem is changing matters as much as how big it is · a real handover names an owner, a first move and when to report back

VERDICT 77W

The fire boundary is the one thing here that gets worse fast enough to cross a line while you are still talking. Fifty-four degrees, climbing six a minute, and the limit is eighty. That is minutes away. The flooding is already losing twenty litres a minute, so it can be watched. The air and the wrist are steady. Everything you hand over needs an owner, a first move and a number that brings it back to you.

TAKEAWAY 16W

A handover is finished when somebody knows what to do first and when to come back.

8.3 Slowing down as a decision

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · BALLPARK

SCENE 29W

The plot is still being rebuilt. The boat makes eight knots toward a bank nine miles ahead. Lieutenant Commander Theo Whitfield asks what coming down to three knots buys.

GUIDE 54W

Four numbers, and two of them are the answers to a smaller question already done. Those are the two trip times. One number is a difference in knots, which is not a difference in hours. Ask which two the comparison needs. Slowing costs distance and buys everything else. The narrows will still be there.

ASK 11W

How much more time does coming down to three knots buy?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

distance divided by speed is time

VERDICT 76W

Time is the only thing in a casualty you can make more of, and speed is the price. Nine miles at eight knots is a little over an hour. The same nine miles at three knots is three hours. Slowing costs progress and buys everything else. The plot gets rebuilt. The flood gets fought. The passage is still there afterwards. Work out what eight knots is spending, in minutes, before deciding the boat can afford it.

TAKEAWAY 10W

Slowing down turns a distance problem into a time budget.

8.4 The trace that will not resolve

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · CALLBACK · CHOICE

SCENE 37W

The fourth trace is faint. It has held nearly the same direction for eleven minutes, and it shows no tone at all on the narrow display. The boat has not changed speed or depth since it appeared.

GUIDE 57W

Two facts have to be explained together. The trace is faint, and it has no tone. Ask of each option whether it covers both. A layer can bend sound and make something faint. It does not pick out tones and remove them. Eleven minutes is a long life for a flicker in software. And nothing aboard changed.

ASK 15W

Machinery makes a steady tone. This trace has none. What does that say about it?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

sound bends when it crosses water at a different temperature · a narrow display looks for steady tones, and a broad one measures sound of every pitch at once

VERDICT 75W

The broad display measures sound of every pitch at once. The narrow one hunts for steady tones. A layer of warmer water can bend a path and make a far-off source arrive faint. It cannot reach in and strip that source's tones away. So faintness and missing tones are two separate clues. The layer explains the first. The second says no tone is loud enough to show. Eleven minutes says it is not a flicker.

TAKEAWAY 15W

Bending can make a sound faint. It cannot pick out and remove that sound's tones.

END OF THE WATCH 11W

Command is deciding what you do, and who does the rest.

The Cooling Path

PLAN CARD 70W

Thirteenth day, and this one was bought two watches ago. A main bearing is running hot. The oil is hot, the fresh water is hot, and the seawater leaving the cooler is barely warmer than the sea going in. The second pump is still shut down from a quiet lineup. This watch you write the cooling path down in order, and find the link that stopped handing the heat on.

BRIEFING 39W

Lube oil temperature is rising on the main bearing and nothing has changed on the throttle. Somewhere between the sea and that bearing there is a chain of things that carry heat away, and one of them is not.

WHAT YOU DECIDE 12W

Trace one cooling path end to end and find where it stopped.

9.1 From the sea to the bearing

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · CHAIN

SCENE 30W

The bearing is running warm at its usual load. Haruki puts the cooling diagram on the table before anybody touches a valve. Six stages, from the bearing to the sea.

GUIDE 62W

Put the five stages in the order the heat actually travels, from the bearing out to the sea. Then read what each handover is doing and name the one that is holding it up. The hottest part is not the answer: heat has to get through every stage, so the chain is limited by the stage that can pass on the least.

ASK 12W

Where does the heat travel, and which handover is holding it up?

BEHIND THE BUTTON 159W

How the heat gets out. The bearing makes heat as it turns. Oil carries it away, a cooler hands it from the oil to seawater, and the seawater carries it overboard. Every one of those is a handover, and if any one of them cannot pass on as much as arrives, the heat backs up behind it.

Why the bearing is the wrong answer even though it is hottest. It is hottest because it is where the heat is made. That would be true even if everything downstream were perfect. What you are looking for is not the hottest stage but the stage that is passing on less than the one before it is delivering.

What the readings tell you. Each link reports what it is doing now — a flow, a temperature difference, a valve position. Compare each one against what it should be handing on, and the stage that has fallen behind is the one to name.

TAKES AS READ

heat has to be handed on through every link before it can leave the boat · the hottest part is not automatically the broken one

VERDICT 79W

The bearing makes the heat, and that does not make the bearing the fault. The heat has to move into the oil, across the oil cooler, into the fresh water, across the sea cooler, and out. Weaken any one of those handovers and everything behind it gets hotter. The readings show normal load and low seawater flow. So the last step out is the one holding it up. Writing the chain first stops the hottest part taking the blame.

TAKEAWAY 16W

A chain of handovers is limited by its weakest required link, not by its hottest part.

9.2 One link is not handing the heat on

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A PERSON · CHOICE

SCENE 33W

The oil and the fresh water are both hot. Seawater goes into the cooler at nine degrees and comes out at twenty-three, against a normal twelve. The flow gauge reads forty-five per cent.

GUIDE 63W

Follow the heat along the chain, and ask where it stops being passed on. Two facts limit the answers. The fresh water is hot, so heat is reaching the cooler. The seawater comes out fourteen degrees warmer instead of three. Ask of each option what it would do to that outlet. A furred cooler would give a small rise, not a big one.

ASK 7W

Which link in the chain has failed?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a cooler needs water moving on both sides to shift heat

VERDICT 79W

How much heat water carries away depends on how much water there is, and on how much it warms up. Keep the heat the same and halve the water, and every litre carries twice as much. So it comes out hotter. Here the sea water warms fourteen degrees instead of three, with the flow at forty-five per cent. The hot fresh water says the heat reaches the cooler. The big rise says too little water is taking it away.

TAKEAWAY 11W

Less water carrying the same heat has to come out hotter.

9.3 The breaker nobody logged

FIRE & ELECTRICAL · ELECTRICAL DISTRIBUTION · A ROOM · CHOICE

SCENE 33W

The second seawater pump did not trip. Its breaker is open at the switchboard. The quiet lineup from two watches ago is still in force, and nobody wrote down what it turned off.

GUIDE 60W

All four options are fair readings of an open breaker. They differ in where the mistake was. Ask of each whether it is about the trade, or about the record of the trade. Shutting the pump was reasonable at the time. What did not survive the watch change was why. The next watch cannot tell a plan from a fault.

ASK 9W

What does this tell you about the quiet lineup?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a watch hands over to people who were not there

VERDICT 74W

The quiet lineup was a deliberate trade, and the reason behind it did not survive the watch change. A log has to carry more than the fact that a machine was shut down. It has to say what that machine was cooling, and when it should come back. Without that, the next watch sees an open breaker and cannot tell a plan from a fault. So this cooling problem is really a handover problem.

TAKEAWAY 11W

A setting nobody wrote down is a setting nobody puts back.

9.4 What the plot is for tonight

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · SPOT

SCENE 25W

Contacts come onto the board through the watch. One standing order says which ones are held, and it is changed as the passage goes on.

GUIDE 32W

Hold what the order asks for and let the rest go. The order changes during the watch and nobody calls it out. What counts is the contacts either side of a change.

ASK 11W

Hold the contacts tonight's order wants, and keep checking the order.

BEHIND THE BUTTON 64W

Why the order changes. In open water the plot is about anything that could close on us. Through a fishing fleet it is about anything under way rather than drifting. Once we go quiet it is about anything that could hear us.

Why the change is the risky part. A contact dropped from the plot leaves nothing behind to say it was ever there.

TAKES AS READ

a watch works to an order about which contacts are held

VERDICT 105W

Three orders run across the watch and each wants different contacts. Anything closing, then anything under way, then anything using active sonar. A contact can match two at once, and that is what makes a change cost something. What the panel scores is the contacts either side of a change. Most of the board is wanted by neither order, so letting those go proves nothing. The control room's own version is the ten minutes after the boat goes quiet. The order has changed to anything that could hear us, and the plot still being kept is the one tracking anything that could run into us.

TAKEAWAY 18W

The cost of an order that has changed is paid by whoever is still using the old one.

9.5 Where the sound will not go

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · CALLBACK · CHOICE

SCENE 34W

The sound trace reads faster above sixty metres and slower below it. A contact held for forty minutes while the boat stayed at twenty. Then it vanished in one sweep, and nothing else changed.

GUIDE 59W

The contact went in one sweep, and nothing else changed. Ask of each option whether it explains that suddenness. Distance and shutting down both work slowly. The blind sector is a direction, not an event. The sound speeds are the numbers doing the work. Sound bends toward slower water, and that can leave a gap our sound never enters.

ASK 14W

The contact was held near the surface and is now gone. Where is it?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

sound travels at different speeds through warmer and colder water · sound bends toward the slower water

VERDICT 79W

Sound does not travel in straight lines through the sea. It bends toward the slower water, and the water below sixty metres here is slower. Bend a path far enough and there are places the sound from our own depth never reaches. A contact that crosses into one of those does not fade away. It goes in a single sweep, while every other trace stays where it was. That suddenness is the shape of a shadow, not of distance.

TAKEAWAY 16W

Sound bends as it travels, which leaves places a listening boat cannot hear into at all.

END OF THE WATCH 14W

A cooling system is a chain, and a chain is diagnosed link by link.

Deep Watch

BEFORE THE WATCH — TAG

Catch Reyes before she goes on watch

Reyes has the trim numbers from last night, and they are on paper in her pocket. Once she is on watch she is at the panel for six hours. The dive planned for this morning needs those numbers first.

PLAN CARD 70W

Twenty-second day, the last full watch, and everything is degraded a little. A contact Rask has not resolved. A plot Sowande has not fixed since morning. A machine nobody has looked at since Tuesday. Whitfield is in the wardroom with a fractured wrist, so the officer you would hand half of this to is gone. This watch you decide the order it all gets worked in, while it is happening.

BRIEFING 33W

An uncertain contact, a plot four hours old, water in the forward bilge, a scrubber train down and a bearing running warm. None of them is an emergency. Together they are the patrol.

WHAT YOU DECIDE 16W

Run the whole boat at once, with everything degraded a little and nothing decided for you.

10.1 Five small problems

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · CHOICE

SCENE 35W

Nothing on the board is red. The plot is four hours old. The contact is faint. The forward bilge is wet, one scrubber is down, and a bearing is warm. The narrows are still ahead.

GUIDE 68W

Nothing on the board is red, so how bad they look will not rank them. Ask of each what is growing: the doubt, the danger, or neither. A faint contact needs time before it can be read. A wet bilge matters if the level moves. One scrubber down is a deadline, not an emergency. The narrows are ahead, and one of these gets worse just by sailing on.

ASK 9W

What is the first thing that has to change?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

how fast something is changing matters more than whether it looks odd

VERDICT 79W

Five small problems are not equally urgent, because they do not change at the same speed. The old plot gets more dangerous with every mile toward the narrows. The faint contact needs time before it means anything. A wet bilge matters when the level moves. A scrubber down is a deadline only if the air is rising. A warm bearing matters if it is climbing. Deal first with the one growing fastest, and give the rest owners and numbers.

TAKEAWAY 17W

A pile of small problems can be worse than one big one, because none of them shouts.

10.2 Everything you are not doing yourself

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A PERSON · PROTOCOL

SCENE 32W

You keep the soft plot yourself, and four other things are open. Whitfield stands by the watch bill with names ready and nobody assigned. This is the last watch of the patrol.

GUIDE 64W

Four open problems and four orders. The pairing is about what each person is meant to notice. Ask of each problem what a useful report would be. A change? A rate? An hour? A cost? A number logged with no threshold is a number nobody acts on. This is the last watch, so an order you have to stand over is not an order.

ASK 10W

Match each open problem to the order that fits it.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

an instruction has to be something the person receiving it can act on alone

VERDICT 74W

A useful order lets somebody act without guessing what you meant. It names who owns it, what to do first, and what brings them back to you. Log the bilge and report a change. Hold the contact until its bearing moves. Restore the scrubber, or name the hour. Get the pump back and say what it costs in noise. Watch it sounds responsible, gives nobody a decision, and leaves every problem attached to you.

TAKEAWAY 15W

An order with a number in it can be acted on alone. Watch it cannot.

10.3 Why the level pulses three times a second

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · BALLPARK

SCENE 32W

Two tones sit at one hundred and forty-seven and one hundred and fifty hertz, on the same bearing. The sound swells and fades in a steady rhythm while both tones hold still.

GUIDE 61W

Three numbers, and one of them is the speed of sound, which the swelling does not need. It belongs to wavelengths, not to this rhythm. Ask of each number whether the swells depend on it. Both tones are holding still, so nothing is changing pitch. What changes is how the two line up, and that comes from the gap between them.

ASK 10W

How many swells a second should those two tones make?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

two sounds arriving together add up, and can also cancel · hertz means how many times a second something repeats

VERDICT 69W

Two tones close together take turns lining up and cancelling. When their crests arrive together the sound is loud. A moment later one has slipped behind, they work against each other, and the sound drops away. How fast that cycle repeats is only the difference between the two pitches. One hundred and fifty against one hundred and forty-seven gives three swells a second. Neither tone has moved at all.

TAKEAWAY 15W

Two close tones together make the loudness rise and fall at the difference between them.

10.4 The one shaft speed that rings

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · CALLBACK · CHOICE

SCENE 29W

At ninety-two turns the after compartment rings loudly enough to hear through a bulkhead. At eighty-five and at a hundred it does not. Nothing else in the plant changes.

GUIDE 65W

All four options are real faults, and three of them get worse as speed goes up. Ask of each whether it could be loud at one speed and quiet either side. Balance problems grow. Bearings get worse. Bubbles start and stay. Only something that has to match a rhythm behaves like this. That decides whether the fix is a repair or a speed to avoid.

ASK 13W

The hull rings at one shaft speed and nowhere else. Why that one?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

everything has speeds at which it prefers to shake, the way a swing does

VERDICT 72W

Everything has a rhythm it prefers to shake at, the way a swing does. Push a swing at any old moment and little happens. Push it in time and it goes higher on every push. The propeller blades hit the water a set number of times each turn. At one shaft speed that beat lands in time with the compartment's own rhythm. Either side of it, the pushes fall out of step.

TAKEAWAY 14W

A small push repeated in time with something's own rhythm builds a large movement.

END OF THE WATCH 14W

Command is choosing which uncertainty you can live with, and saying so out loud.

THE CLOSING CARD

The boat surfaced at dawn on the twenty-third day, into weather nobody had seen for three weeks. The fractured wrist is in a cast and signed fit for the transit. The contact was resolved, the plot was fixed, and the machine that had been left since Tuesday was opened up and found in time.

Nothing was lost on this patrol that could not be repaired. The whole crew went up the ladder into daylight, which is the only measure of a watch that matters.

Card text only — no options, boards or answers. 8,107 words of card prose across 41 stops. Generated from themes/deepwatch_ms by tools/make-cardbook.mjs.